Math 121: Commutative Algebra

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Problem 1 (5 points). Let $\varphi : R \to S$ be a ring homomorphism. Carefully prove that the induced map on spectra

$$\varphi^{\#} : \mathrm{Spec}(S) \to \mathrm{Spec}(R)$$

is continuous in the Zariski topology.

Solution.

Problem 2 (5 points). Let R be a ring and let $f_1, f_2, \ldots, f_t \in R$ be non-zero elements such that $\{D(f_i)\}_{i=1}^{t}$ form an open cover of Spec(R). Carefully prove that R is reduced if and only if $R\left[\frac{1}{f_i}\right]$ is reduced for all i.

Solution.